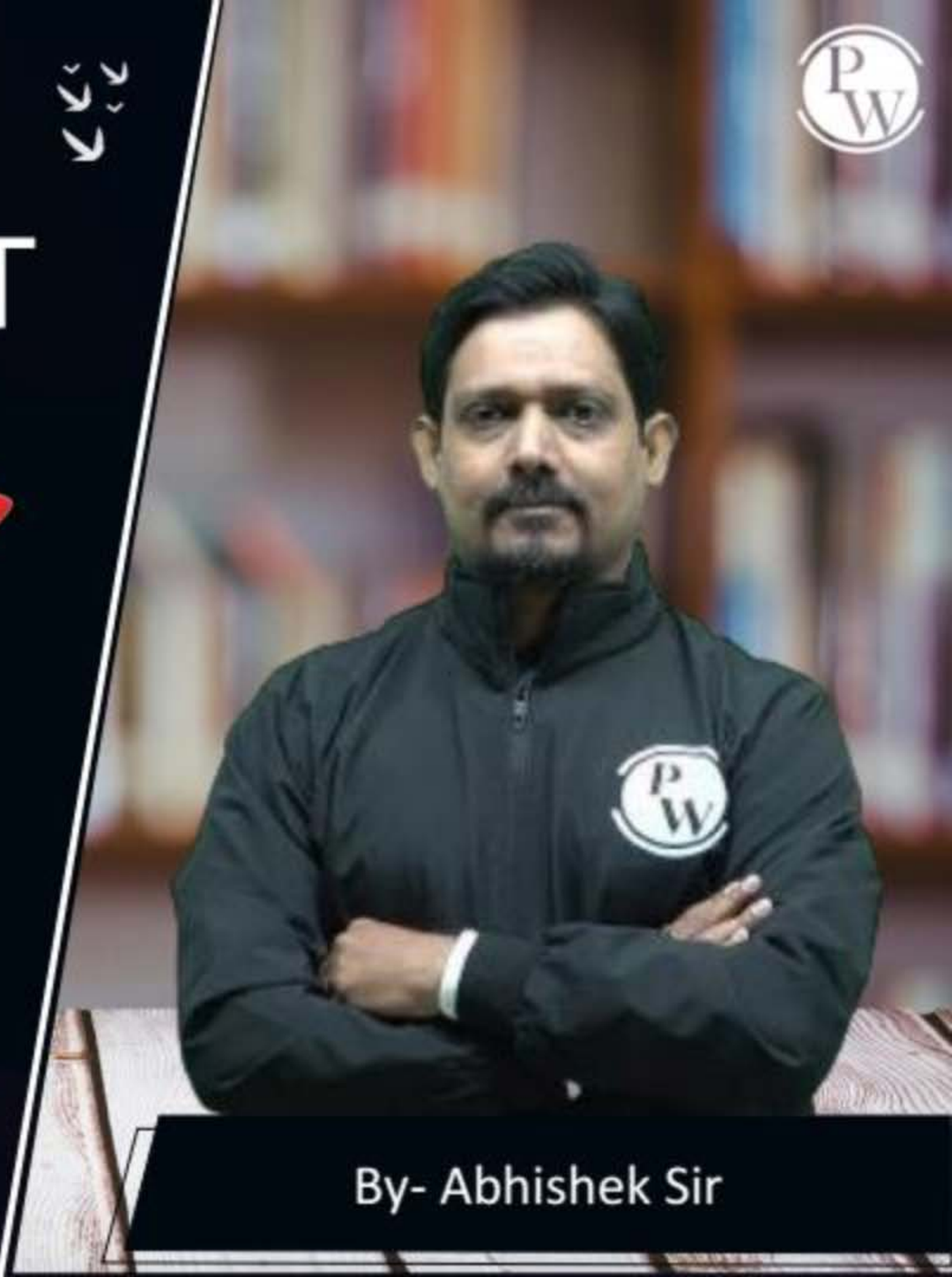
Computer Science & IT



C Programming

Array & Pointers

Lecture No. 04



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Recap of Previous Lecture



- Topic — One Dimensional Array
- Topic
- Topic
- Topic
- Topic

pointer Arithmetic

$a[i] = *(a+i) = *(i+a)$

$= i[a]$

$a[0] = 0[a]$ ← allowed

Slide

Topics to be Covered



- Topic: Two Dimensional Array
- Topic
- Topic
- Topic
- Topic

Slide



Array

C - Languag there is No provision for restriching array out of bound. ← Its responsibility of programmar that it Access the element within boundary

ANSI C -

Slide



Address of an Array

int a[] = {1, 2, 3, 4};

a ← Name address of first element of array

integer

a represent Address of an integer

* with Constant can be added or Substracted.
* ++ and --
* Two similar pointer can be Substracted

Slide



Array

```c
#include <stdio.h>

int main(){
    int a[] = {11,12,13,14 };
    printf("%p\n",a );
    printf("%p\n",&a );
    printf("%p\n",&a[0]);
    return 0;
}
```

A[0]	A[1]	A[2]	A[3]
11	12	13	14
100	104	108	112

%u

a address of first element of array

Imp & a : Address of one dimensional

① will print 100 (integer address)

② will print 100 (array of integer)

%p pointer

Hexadecimal

Slide

1-D

Both array address

first integer address

will be same

&a[0]

&(*(a+0))

&(*100)

&(11)

= 100

> & Name of array
>
> is Address of array type



Array

```c
#include <stdio.h>

int main(){
    int a[] = {11,12,13,14 };
    printf("%p\n",a );
    printf("%p\n",&a );
    printf("%p\n",&a[0]);
    printf("%p\n",a+1 );
    printf("%p\n",&a+1 );
    printf("%p\n",&a[0]+1);
    return 0;
}
```

A[0]	A[1]	A[2]	A[3]
11	12	13	14
100	104	108	112

100 integer

100 Array (Address)

100 - integer

(100+1) = 100+1×4 = 104

(100+1) = 100+1×16 = 116

(100+1) = 100+1×4 = 104

Array ka address

&a+1 jump size of array

4×4 = 16

Slide



Arithmetic Operation with Pointer

#Q Assume that array elements are stored from 1000, 1004, 1008, 1012, 1016

```
int main (){
  int a[] ={10,20,30,40,50};
  int i,*b,*b1;
  b = a;          //1000 is assigned
  b1 = a+4        //1000 +4 ×4 = 1016 assigned
  i = b1-b;
  printf("%d",i);
  return 0;
}
```

The value printed by the program is_____?

b = a

Question

$b_1 = a + 4$

↑ integer

$a + 4 \times 4$

$1000 + 16 = 1016$

$i = (1016 - 1000) = 16$

$= \frac{16}{\text{Size}} = \frac{16}{4} = 4$

ptr$_1$ & ptr$_2$ is pointer of Similar kind

$$ptr1 - ptr_2 = \frac{ptr_1 - ptr_2}{\text{Size of data type}}$$

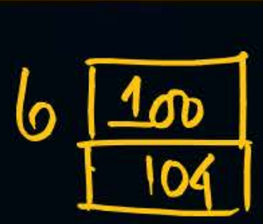
$$ptr_1 = ptr_2 + 4$$

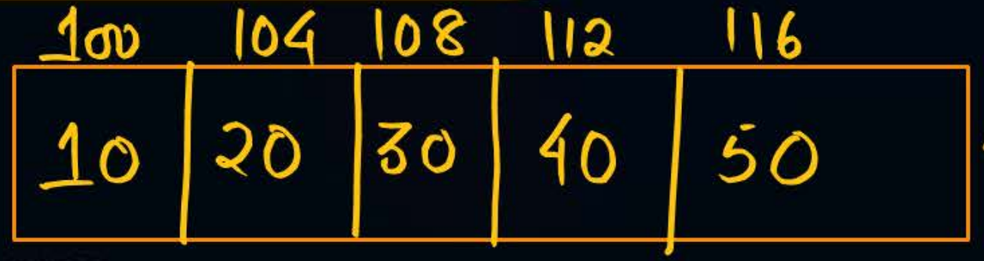
$$ptr_1 = ptr_2 + 4 \times \text{Size}$$

$$ptr_1 - ptr_2 = 4 \times \text{Size} \Rightarrow 4 = \frac{Ptr - ptr_2}{\text{Size}}$$



Arithmetic Operation with Pointer





a[0] a[1] a[2] a[3] a[4] &a[4]

```
#Q
main (){
      int a[] ={10,20,30,40,50};
      int i,*b;
      b = &a[4]-4;
      for(i=0;i<=4;i++){
            printf("%d",*b);
            b++;
      }
return 0;
}
Output_____
```

b++; → b = b+1

b = 116 - 4 = 116 - 4*4 = 116 - 16 = 100

	*b	b++
i=0	10	104
i=1	20	108
i=2	30	112
i=3	40	116
i=4	50	120

10,20,30,40,50

Answer.

Slide

Arithmetic Operation with Pointer

&(a+3)

(a+3)

```
#Q
 include<stdio.h>
int main()
{
int a[] = {1,23,4,5};
int *b= a+3;

printf("%d ", b[-2]);
}
```

(A) 1
(B) 23
(C) 4
(D) 5

1	23	4	5
100	104	108	112

int *b = a+3

100+3 = 100+3×4 = 112

b[-2] = *(b-2)

*(112-2) = *(112-2×4)

*(112-8) = *(104) = 23

Slide



Arithmetic Operation with Pointer

#Q

What is the output of the following program ?

```
int i , b [] = {2, 3, 4, 5, 6}, * p ;
p = b ;
++ * p ;
printf ("& d" ,*p) ;
p + = 2 ;
printf (" & d " , *p)
```

(A) 1 3

(B) 2 3

(C) 2 4

(D) 3, 4

Slide

2 D Array Declaration & Initialization

```
int a[2][3] =
{{1,2,3},{4,5,6}};
```

Row 0 ⟶

Row 1 ⟶

Column 0	Column 1	Column 2
a[0][0] 1	a[0][1] 2	a[0][2] 3
a[1][0] 4	a[1][1] 5	a[1][2] 6

a[1][1]

Logical diagram

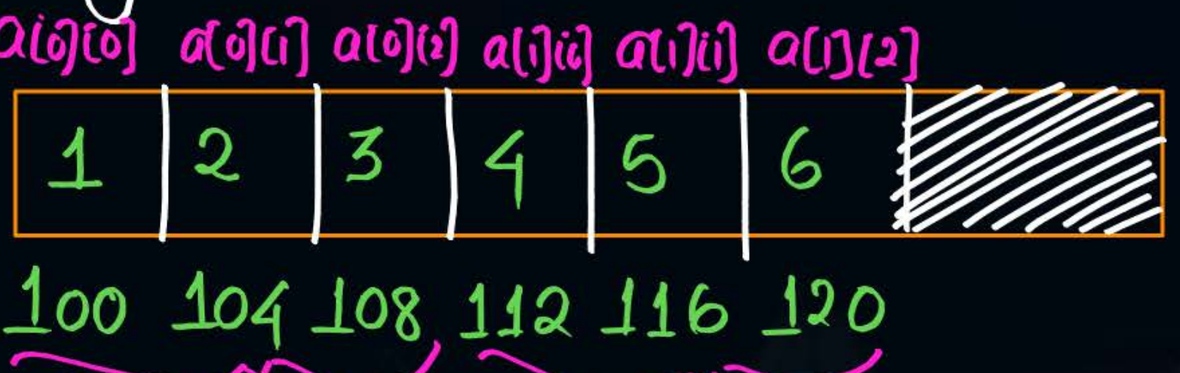
Index starts from 0 for Row & column both

Slide

2 D Array Declaration & Initialization

2 D array stored in memory in Sequential manner.

```
int a[2][3] = { {1,2,3},
                {4,5,6} };
```



In 2 D array Row are stored one after another.

Question: each element of 2 D array is? 1 D array.

Slide



2 D Array Declaration & Initialization

int a[2][3] = { {1,2,3},
{4,5,6} };

a ←

*a

**a

2a

integer

a

{ {1,2,3},
{4,5,6} };

a[i][j]

*(a+i)[j]

((a+i)+j)

Slide

1. a. Represent Address of first element of array

Each element of 2D array is 1D array.

a. array data type. (1D)

2. *a Represent address first element of 1D array

*a represent Address of integer.

3 **a represent an integer ✓

4 &a An address of 2D array



Meaning A[i][j]

$$a[i][j] = *(a+i)[j]$$

$$= *(*(a+i)+j)$$

Two pointers required to access element 2D array

2-D Array

```
#include <stdio.h>

int main(){
    int a[2][2] = {{11,12},{13,14}};

    printf("\n %u",a);
    printf("\n %u",&a);
    printf("\n %u",*a);
    printf("\n %u",**a);

    printf("\n %u",a+1);
    printf("\n %u",*a+1);
    printf("\n %u",**a+1);
    printf("\n %u",&a+1);

    return 0;
}
```

108
104
12
116

a[0][0] a[0][1] a[1][0] a[1][1]

{ {11 | 12} | {13 | 14} }

100 104 108 112

Row-0 Row-1

a Address of 1D array - 100

2a Address of 2D array - 100

*a Address of an integer : 100

**a - integer = (11)

Slide



2 mins Summary

Topic

Two Dimensional Array

Topic

Topic

Topic

Topic

THANK - YOU

